Spatial Operations

Alexander Abuabara

DAEN 489 ISEN 489 Spring 2026

ENGINEERING
TEXAS A&M UNIVERSITY

Overview

Vector Operations

- Spatial subsetting
- Topological relations
- Distance calculations
- Spatial joining
- Spatial aggregation

Raster Operations

- Spatial subsetting
- Map algebra
- Local operations
- Focal operations
- Zonal operations

Learning Objectives

- Apply spatial subsetting, joins, and aggregation on vectors
- Perform local, focal, zonal, and global raster operations
- Differentiate key spatial operations on vector and raster data
- Interpret spatial relationships (intersects, within, disjoint)

Practical Applications

- **Urban Planning**:
  Points in polygon analysis, buffer zones, accessibility studies
- **Environmental Analysis**:
  Habitat suitability, watershed analysis, land cover change
- **Remote Sensing**:
  NDVI calculation, image classification, change detection
- **Emergency Response**:
  Service area analysis, resource allocation, risk assessment
- **Transportation**:
  Network analysis, route optimization, catchment areas

Vector Spatial Operations

Vector Spatial Operations

- Many spatial operations have a non-spatial (attribute) equivalent
  - So, concepts such as subsetting and joining datasets (demonstrated last week) are applicable here
  - This is especially true for vector operations
- Spatial operations differ from non-spatial operations in several ways
  - However spatial joins, for example, can be done in several ways while the attribution joins can only be done in one way.

Spatial Subsetting

- Selecting features based on spatial relationships
  - Process of taking a spatial object and returning only features that relate in space to another object
- Similar to attribute subsetting but uses spatial relations
- Can specify alternative operators
- Default relation: `st_intersects()`

R Syntax

```
# Basic subsetting
x[y, ]

# With specific operator
x[y, , op = st_within]

# Using st_filter()
x |> st_filter(y)
```

Topological Relations

Binary predicates describing spatial relationships

`st_intersects()`	Features that touch, cross, or are within
`st_touches()`	Features that share a boundary
`st_overlaps()`	Features that partially overlap
`st_within()`	Features completely inside another
`st_crosses()`	Features that cross each other
`st_disjoint()`	Features with no spatial relationship

Distance Relations

`st_distance()`

- Calculates continuous distance between spatial objects
- Returns values with units (meters, feet, etc.)
- Can compute distance matrices
- Supports spherical geometries (S2)
- Distance to any part of polygon

Example Usage

```
# Single distance
st_distance(point_a, point_b)

# Distance matrix
st_distance(points, polygons)

# Within distance
st_is_within_distance(
  x, y, dist = 20
)
```

Spatial Joins

`st_join()`

- Adds columns from source object to target based on spatial overlap

Types of Joins

- **Left join** (default): keeps all target features
- **Inner join**: only matching features
- **Distance-based**: joins within threshold

Syntax

```
# Basic spatial join
st_join(target, source)

# With custom predicate
st_join(x, y,
        join = st_within)

# Distance-based
st_join(x, y,
  st_is_within_distance,
  dist = 100)
```

Spatial Aggregation

- Condenses data using aggregating functions (mean, sum, etc.) based on the geometry of a grouping object

Two approaches

- **Base R**, concise syntax

```
aggregate(x, by = y, FUN = mean)
```

- **tidyverse**, more flexible

```
st_join() |> group_by() |> summarize()
```

Raster Operations

Map Algebra and Spatial Processing

Map Algebra

Operations that modify or summarize raster cell values

Local	Cell-by-cell operations	Addition, subtraction, classification, Normalized Difference Vegetation Index (NDVI)
Focal	Neighborhood operations (3×3 window)	Smoothing, edge detection, slope
Zonal	Operations by categorical zones	Statistics by land cover type
Global	Entire raster operations	Min, max, mean, distance calculations

Local Operations

- Cell-by-cell operations on one or more rasters

Common Operations

- **Arithmetic**: +, -, *, /
- **Logical**: >, <, ==
- **Mathematical**: log(), sqrt(), ^2
- **Classification**: reclassify values
- **Multi-layer**: NDVI, band math

NDVI Example

```
# Define NDVI function
ndvi_fun = function(nir, red) {
  (nir - red) / (nir + red)
}

# Apply to raster layers
ndvi = lapp(
  multi_rast[[c(4, 3)]],
  fun = ndvi_fun
)
```

Focal Operations

`focal()`

- Neighborhood-based operations using moving windows
- Applies aggregation function to neighborhoods (kernels)

Applications

- **Smoothing filters** (low-pass)
- **Edge detection** (high-pass)
- **Terrain analysis** (slope, aspect)
- **Image processing**

Example

```
# Define 3×3 window
w = matrix(1, nrow=3, ncol=3)

# Apply minimum function
focal(raster,
      w = w,
      fun = min)

# Terrain processing
terrain(dem, v='slope')
```

Zonal Operations

`zonal()`

- Aggregation by categorical zones
- Computes statistics for cells grouped by zones defined in a second raster

Key Differences from Focal

- Zones **don't** have to be neighbors
- Defined by **categorical** raster
- Returns **summary table** by zone
- Can return raster with **`as.raster=TRUE`**

Example

```
# Calculate mean elevation by zone
z = zonal(
  elev,
  zones,
  fun = 'mean'
)
```

Key Considerations

Coordinate Reference Systems

- Both objects must have same CRS
- Reproject if necessary before operations
- Critical for accurate results

Spatial Indexing

- Uses Sort-Tile-Recursive (STR) algorithm
- Speeds up spatial queries significantly
- Automatic in `st_join()` and predicates

Memory Management

- Sparse matrices for efficient storage
- Set `sparse=FALSE` for dense output
- Consider aggregation for large datasets

Next Week

Spatial Data Operations on Vector Data

- Simplification
- Centroids
- Buffers
- Clipping
- Geometry unions

Spatial Data Operations on Raster Data

- Extent and origin
- Aggregation and disaggregation
- Resampling

Practice Question 1

Subdivision Parcel Analysis (Vector)

- Find spatial datasets for Brazos County:
  - Polygon layer containing subdivision boundaries with fields such as *subdiv_id*, *subdiv_name*
  - Point or polygon layer containing all property parcels with fields such as *parcel_id*, *land_value*, *improvement_value*, *property_type*
- Filter the parcels dataset to include only residential ones
- Perform a spatial join to estimate the residential parcels within each subdivision
  - Calculate the average land and improvement values for each subdivision
- Identify which subdivision has:
  - The highest number of residential parcels
  - The highest average land value per parcel
  - The highest average improvement value per parcel

Practice Question 2

Elevation Analysis (Raster)

- Find the Digital Elevation Model (DEM)
- Load individual DEM tiles and visualize
- Check the number of layers (bands), spatial res., CRS, extent (bounding box)
- Merge tiles into a single DEM (using the appropriate merging technique)
- Plot the “mosaicked” DEM (and save it as a new raster)
- Calculate the following terrain derivatives:
    - Slope, Aspect, Terrain Ruggedness Index (TRI), Topographic Position Index (TPI)
- Apply a 3×3 focal filter to smooth the DEM (use the min, max, and mean functions)
- Generate histograms for elevation (and slope)
    - Calculate summary statistics (min, max, mean, standard deviation)

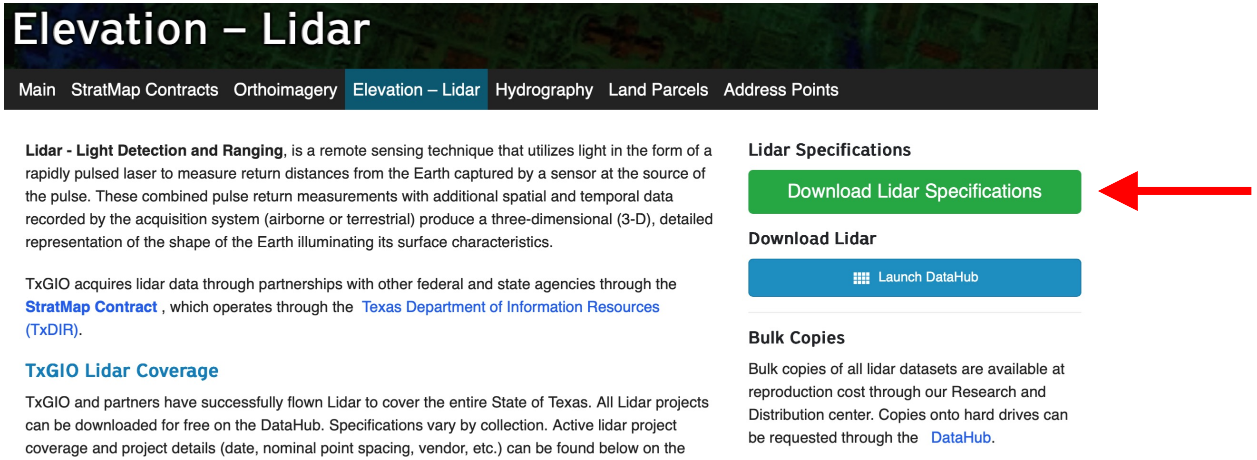
Practice Data (Hint)

Vector

- Parcel: https://brazoscad.org/tax-information/gis/

Raster

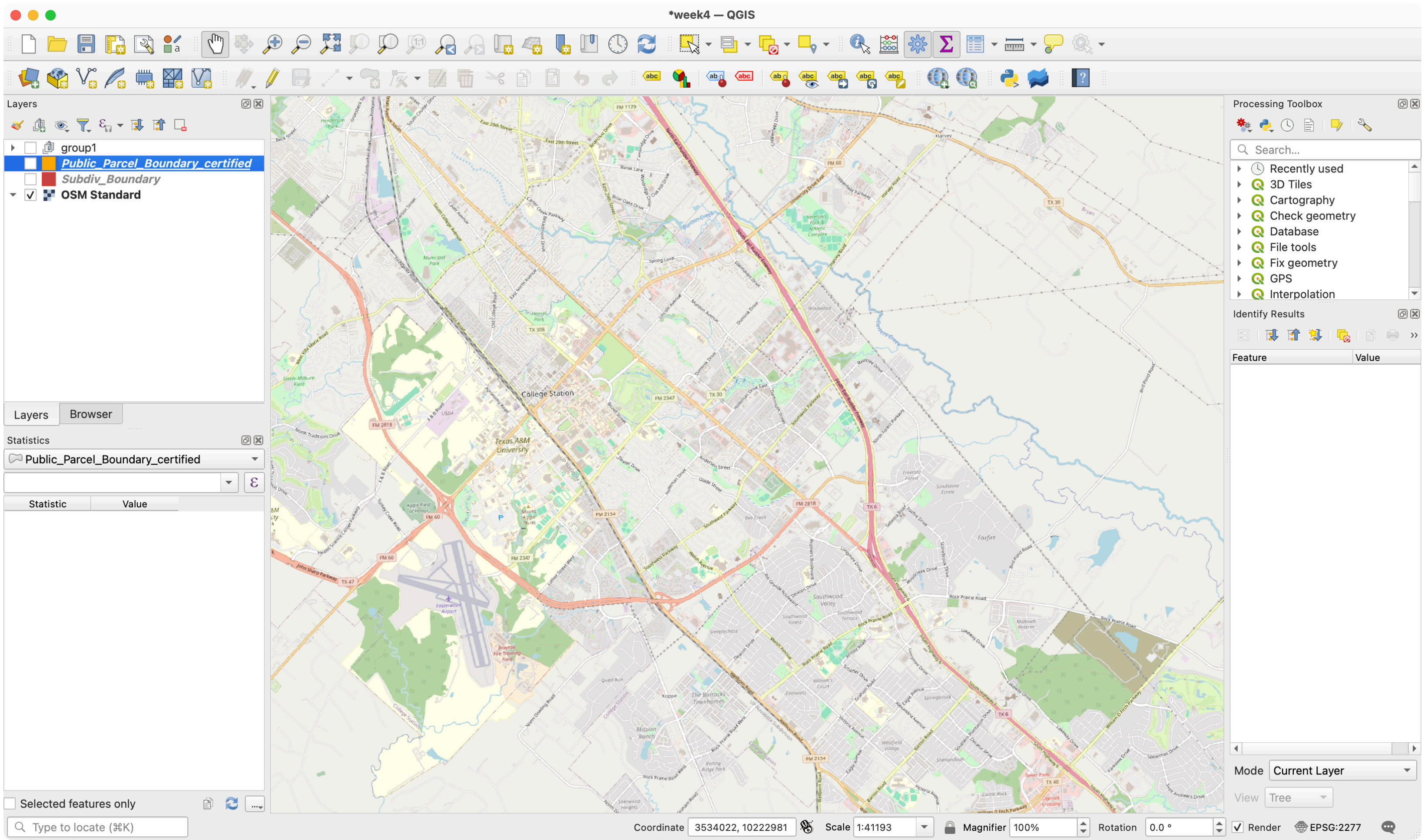
- Elevation: https://tnris.org/stratmap/elevation-lidar.html
  - Note: Lidar Specs https://geographic.texas.gov/_uploads/TxGIO_Lidar_Spec_v14.pdf

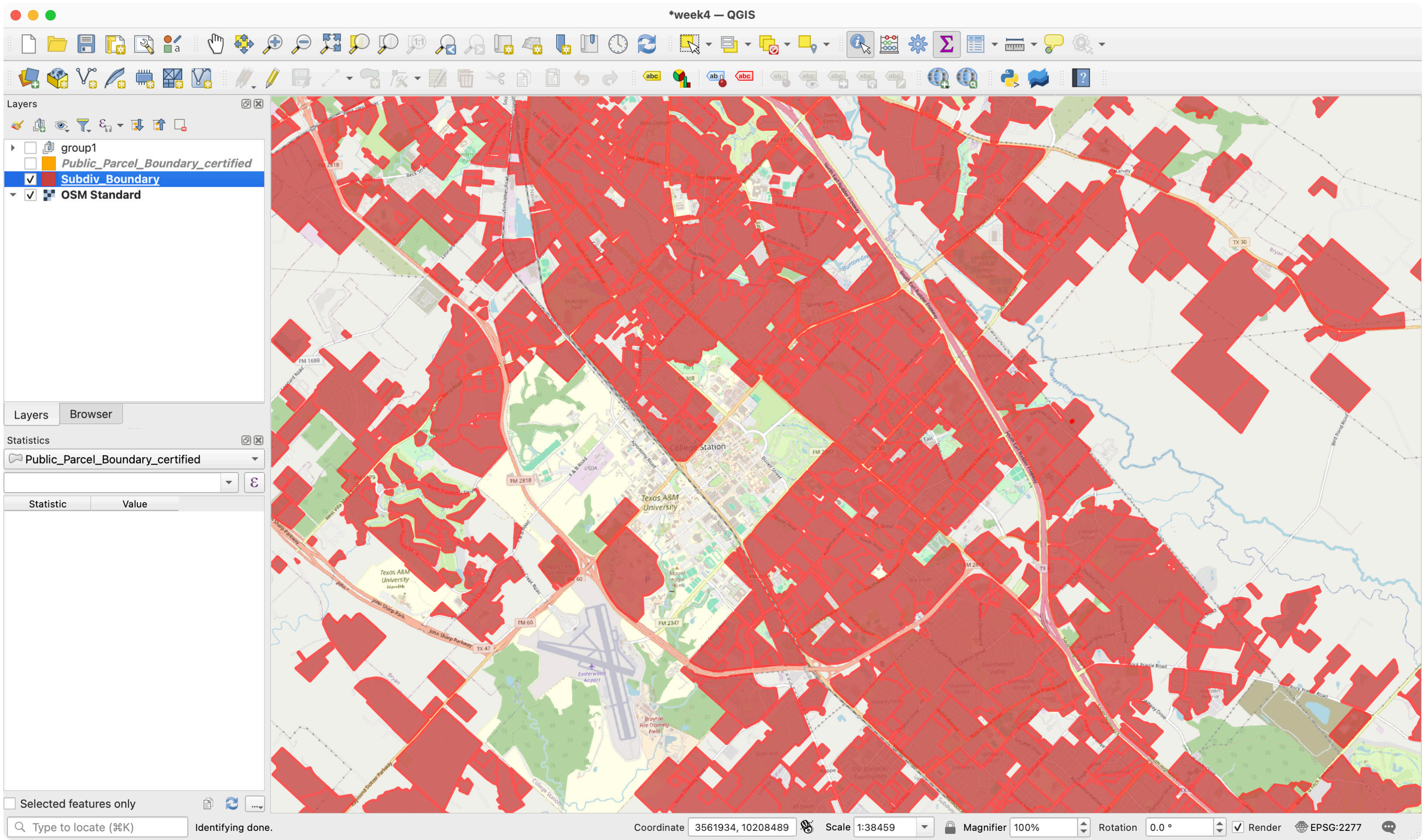


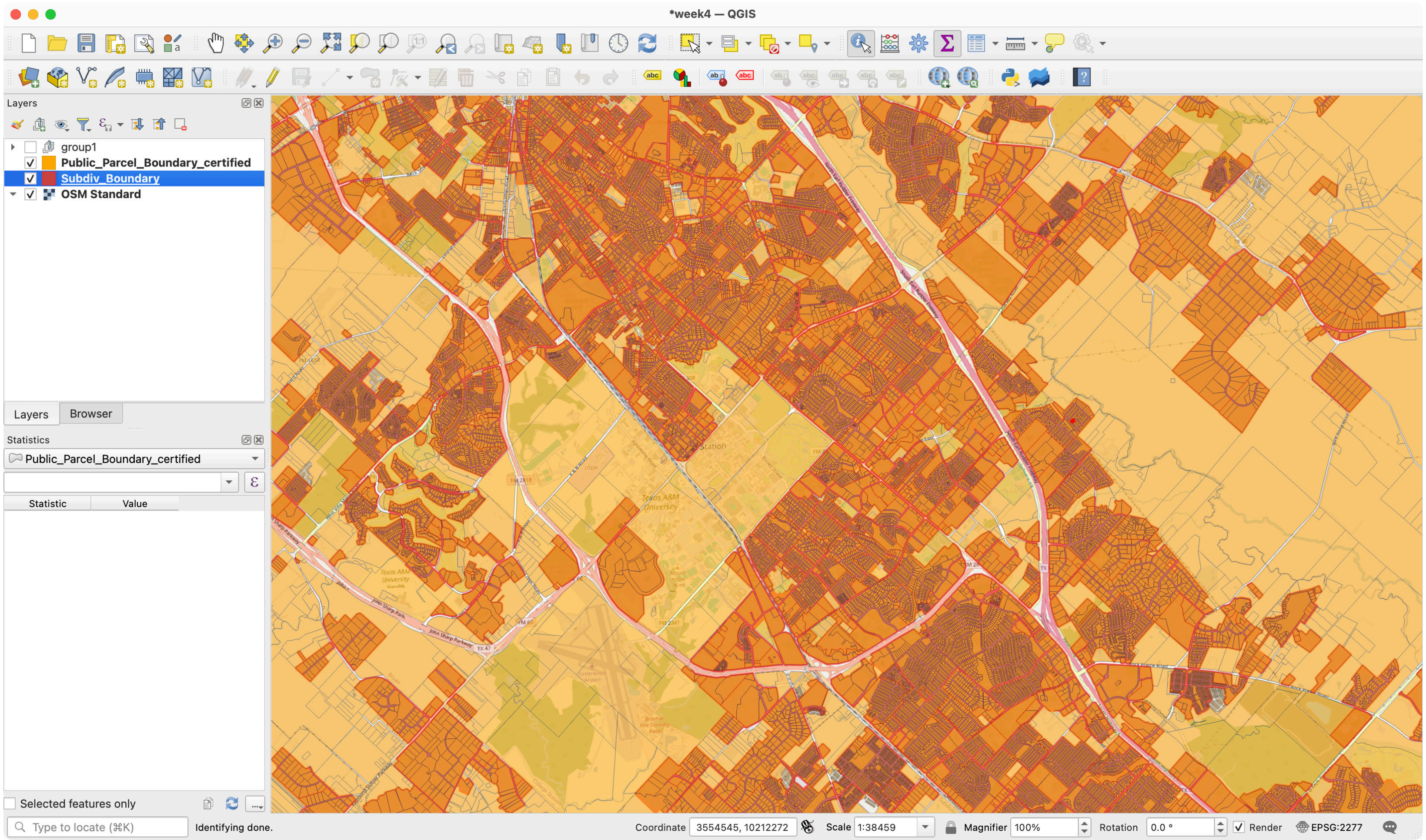
Script

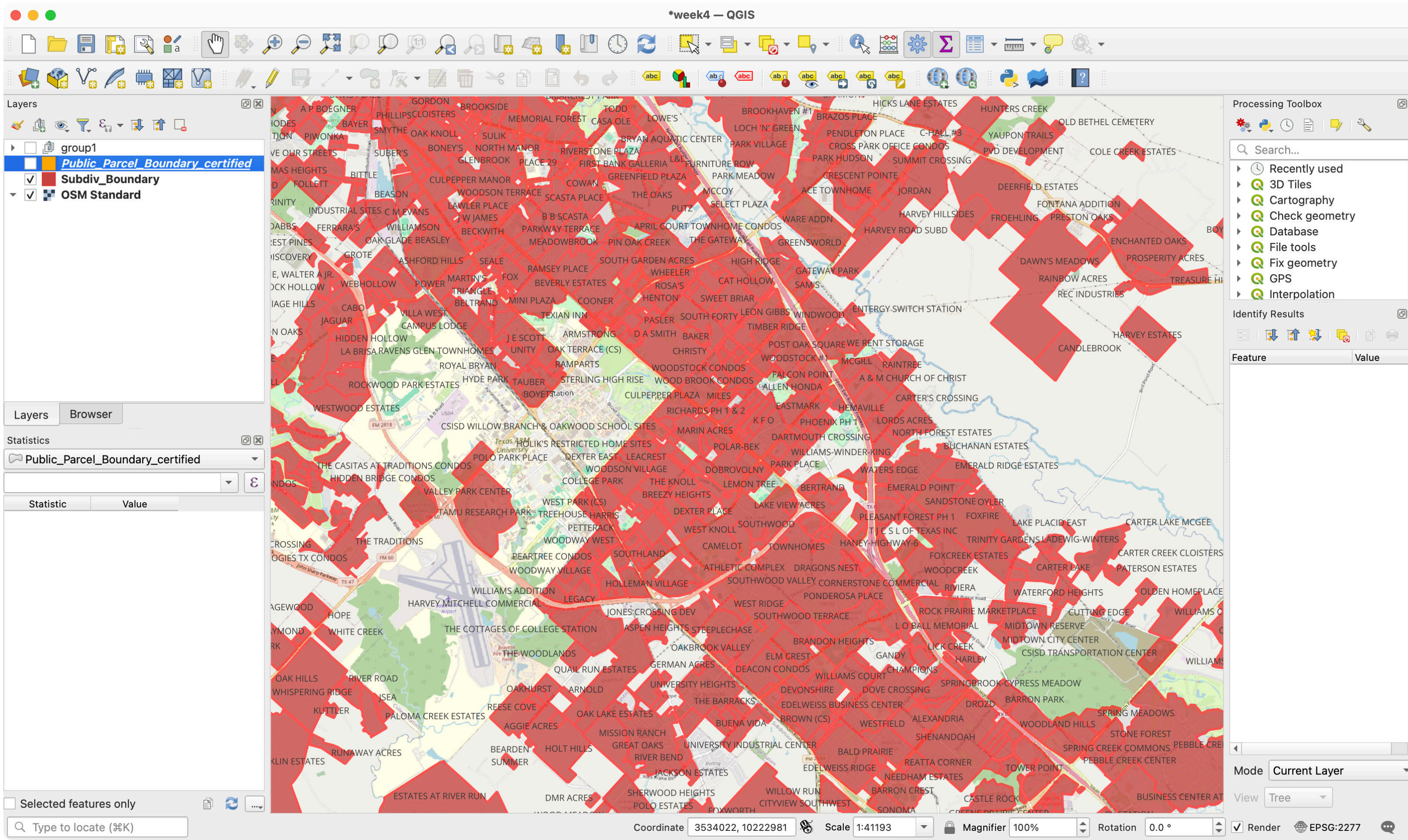
- After the in-class practice, compare your work with https://abuabara.github.io/DAEN489S26/week4.r

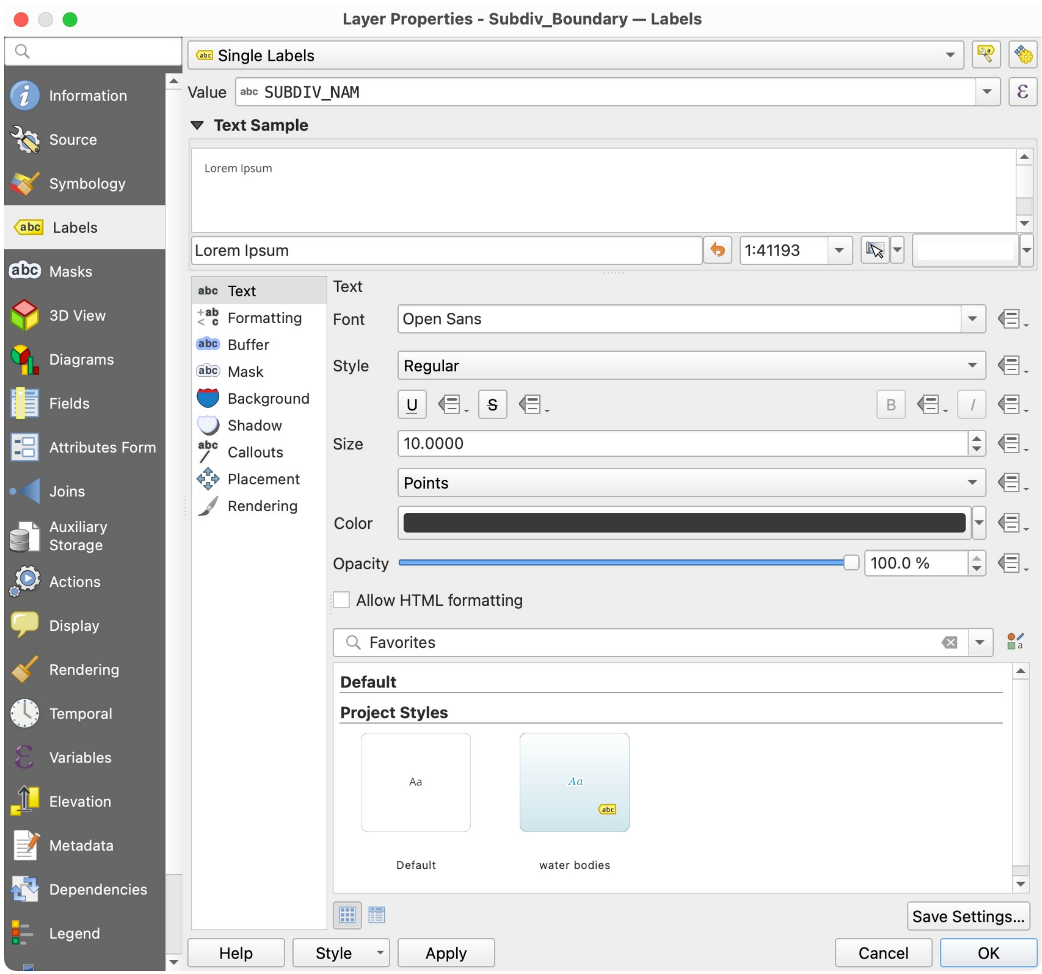
QGIS

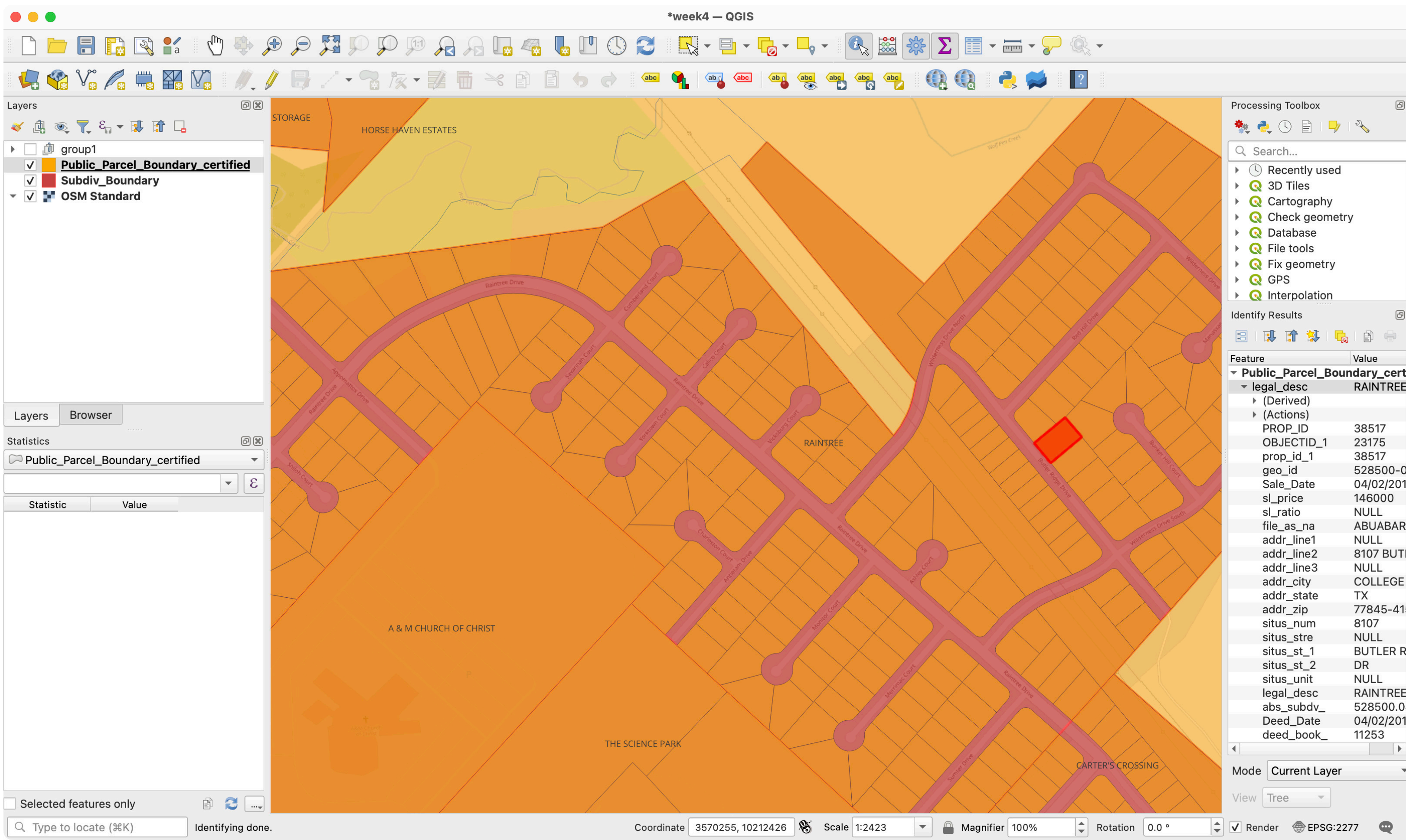

*week4 — QGIS
Layers
group1
Public_Parcel_Boundary_certified
Subdiv_Boundary
OSM Standard
Layers
Browser
Statistics
Public_Parcel_Boundary_certified
Statistic
Value
Selected features only
Type to locate (⌘K)
Processing Toolbox
Search...
Recently used
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Identify Results
Feature
Value
Mode
Current Layer
View
Tree
Coordinate
3534022, 10222981
Scale
1:41193
Magnifier
100%
Rotation
0.0 °
Render
EPSG:2277
College Station
Texas A&M University

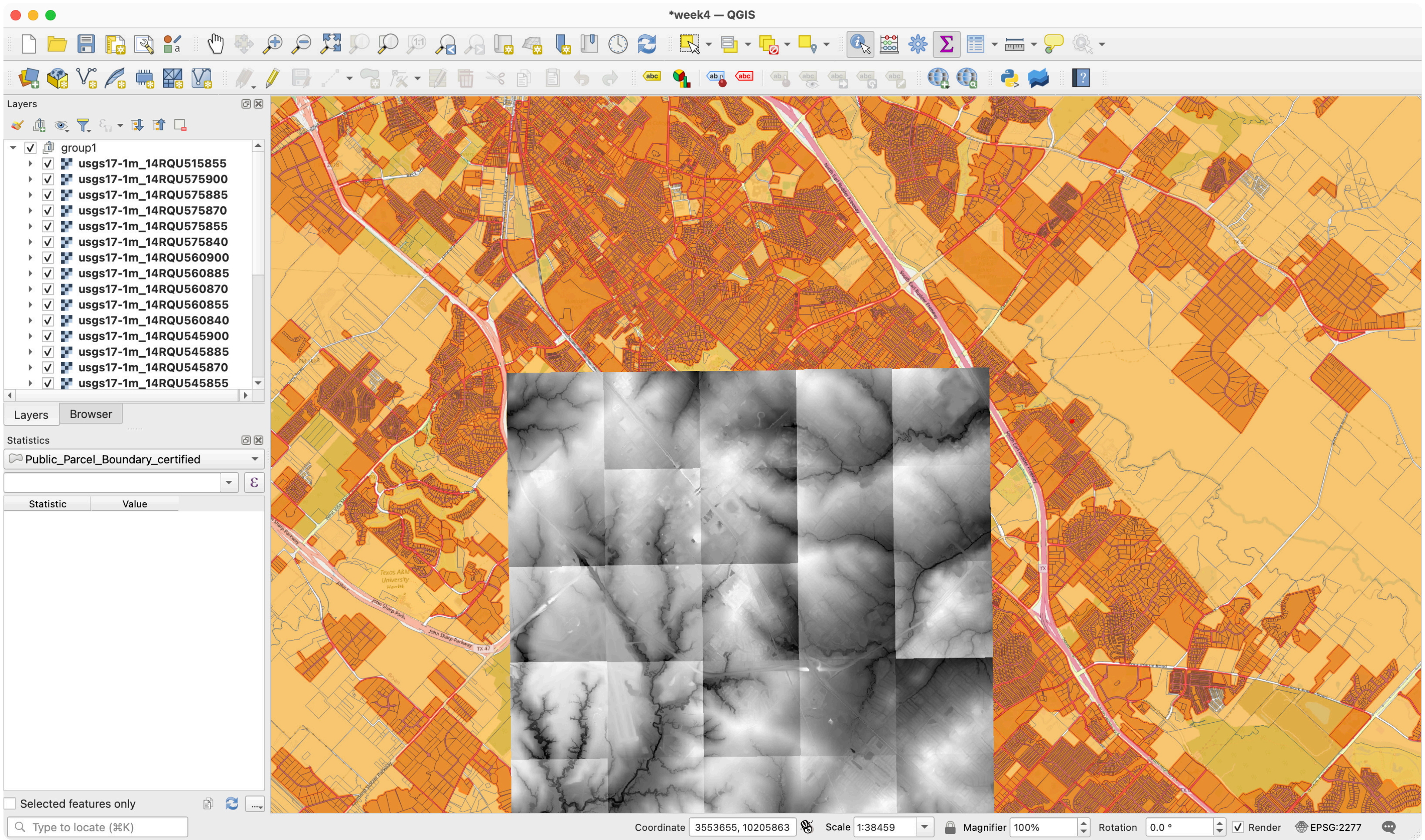

*week4 — QGIS
Layers
group1
Public_Parcel_Boundary_certified
Subdiv_Boundary
OSM Standard
Layers
Browser
Statistics
Public_Parcel_Boundary_certified
Statistic
Value
Selected features only
Type to locate (⌘K)
Identifying done.
Coordinate
3561934, 10208489
Scale
1:38459
Magnifier
100%
Rotation
0.0 °
Render
EPSG:2277
College Station
Texas A&M University


*week4 — QGIS
Layers
group1
Public_Parcel_Boundary_certified
Subdiv_Boundary
OSM Standard
Layers
Browser
Statistics
Public_Parcel_Boundary_certified
Statistic
Value
Selected features only
Type to locate (⌘K)
Identifying done.
Coordinate
3554545, 10212272
Scale
1:38459
Magnifier
100%
Rotation
0.0 °
Render
EPSG:2277


*week4 — QGIS
Layers
group1
Public_Parcel_Boundary_certified
Subdiv_Boundary
OSM Standard
Layers
Browser
Statistics
Public_Parcel_Boundary_certified
Statistic
Value
Selected features only
Type to locate (⌘K)
Processing Toolbox
Search...
Recently used
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Identify Results
Feature
Value
Mode
Current Layer
View
Tree
Coordinate
3534022, 10222981
Scale
1:41193
Magnifier
100%
Rotation
0.0 °
Render
EPSG:2277


Layer Properties - Subdiv_Boundary — Labels
Information
Source
Symbology
Labels
Masks
3D View
Diagrams
Fields
Attributes Form
Joins
Auxiliary Storage
Actions
Display
Rendering
Temporal
Variables
Elevation
Metadata
Dependencies
Legend
Single Labels
Value
SUBDIV_NAM
Text Sample
Lorem Ipsum
Lorem Ipsum
1:41193
Text
Formatting
Buffer
Mask
Background
Shadow
Callouts
Placement
Rendering
Text
Font
Open Sans
Style
Regular
Size
10.0000
Points
Color
Opacity
100.0 %
Allow HTML formatting
Favorites
Default
Project Styles
Default
water bodies
Save Settings...
Help
Style
Apply
Cancel
OK


*week4 — QGIS
Layers
group1
Public_Parcel_Boundary_certified
Subdiv_Boundary
OSM Standard
Layers
Browser
Statistics
Public_Parcel_Boundary_certified
Statistic
Value
Selected features only
Type to locate (⌘K)
Identifying done.
STORAGE
HORSE HAVEN ESTATES
RAINTREE
A & M CHURCH OF CHRIST
THE SCIENCE PARK
CARTER'S CROSSING
Processing Toolbox
Search...
Recently used
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Identify Results
Feature
Value
Public_Parcel_Boundary_cert
legal_desc RAINTREE
(Derived)
(Actions)
PROP_ID 38517
OBJECTID_1 23175
prop_id_1 38517
geo_id 528500-0
Sale_Date 04/02/201
sl_price 146000
sl_ratio NULL
file_as_na ABUABAR
addr_line1 NULL
addr_line2 8107 BUTI
addr_line3 NULL
addr_city COLLEGE
addr_state TX
addr_zip 77845-41
situs_num 8107
situs_stre NULL
situs_st_1 BUTLER R
situs_st_2 DR
situs_unit NULL
legal_desc RAINTREE
abs_subdv_ 528500.0
Deed_Date 04/02/201
deed_book_ 11253
Mode Current Layer
View Tree
Coordinate 3570255, 10212426
Scale 1:2423
Magnifier 100%
Rotation 0.0 °
Render
EPSG:2277


*week4 — QGIS
Layers
group1
usgs17-1m_14RQU515855
usgs17-1m_14RQU575900
usgs17-1m_14RQU575885
usgs17-1m_14RQU575870
usgs17-1m_14RQU575855
usgs17-1m_14RQU575840
usgs17-1m_14RQU560900
usgs17-1m_14RQU560885
usgs17-1m_14RQU560870
usgs17-1m_14RQU560855
usgs17-1m_14RQU560840
usgs17-1m_14RQU545900
usgs17-1m_14RQU545885
usgs17-1m_14RQU545870
usgs17-1m_14RQU545855
Layers
Browser
Statistics
Public_Parcel_Boundary_certified
Statistic
Value
Selected features only
Type to locate (⌘K)
Coordinate
3553655, 10205863
Scale
1:38459
Magnifier
100%
Rotation
0.0 °
Render
EPSG:2277